Prisilni odziv na cos

$\boxed{3^\circ}$ specijalni slučaj (nastavak na 6.10)

→ uvrštavamo u $y = \dfrac{b_0 s^N + b_1 s^{N-1} + \dots + b_N}{s^N + a_1 s^{N-1} + \dots + a_N} \cdot u$

▸ $s = -j\omega$

$u(t) = U e^{-j\omega t}$

$$\Rightarrow H(-j\omega) = \frac{b_0 (-j\omega)^N + b_1 (-j\omega)^{N-1} + \dots + b_N}{(-j\omega)^N + a_1 (-j\omega)^{N-1} + \dots + a_N}$$

$$y(t) = H(-j\omega) \cdot U e^{-j\omega}$$

* u odnosu na 2. slučaj ($s = j\omega$) u kojem je $y(t) = H(j\omega) U e^{j\omega}$

ovo je konjugirano kompleks rješenje.

$\boxed{4^\circ}$ $u(t) = U\cos(\omega t) = U \dfrac{e^{j\omega t} + e^{-j\omega t}}{2} = \dfrac{U}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$

$$y_p(t) = H(j\omega)\frac{U}{2} e^{j\omega t} + H(-j\omega)\frac{U}{2} e^{-j\omega t} = \left(\frac{U \cdot H(j\omega) e^{j\omega t}}{2}\right) + \underbrace{\left(\frac{U \cdot H(j\omega) \cdot e^{j\omega t}}{2}\right)^*}_{\text{konjugirano kompl.}}$$

→ njihovi realni dj su jednaki
imaginarni isti ali suprotnog predznaka

$\Rightarrow$ ostaje samo realan dio: $y_p(t) = \cancel{2} \cdot \mathrm{Re}\left\{\dfrac{H(j\omega) e^{j\omega t}}{\cancel{2}} \cdot U\right\}$

$$y_p(t) = \mathrm{Re}\left\{|H(j\omega)| e^{j\measuredangle H(j\omega)} \cdot U e^{j\omega t}\right\} = |H(j\omega)| \cdot U \cdot \cos(\omega t + \measuredangle H(j\omega))$$

izraz je krenuo od vremenske exp pa je i kosinus vremenski

Primjer: onaj koji smo pokazivali: $y''(t)+2y'(t)+2y(t)=u(t)$

$u(t)=3\cos(2t)$

$\downarrow$

$\omega=2$

$$* \; H(j\omega)=\frac{1}{(j\omega)^2+2(j\omega)+2}$$

$$|H(j\omega)|=\frac{1}{\sqrt{\omega^4+4}} \qquad \measuredangle H(j\omega)=-\operatorname{arctg}\frac{2\omega}{2-\omega^2}$$

$$|H(j\omega)|=\frac{1}{\sqrt{2^4+4}}=\frac{1}{\sqrt{20}}=0{,}2236$$

$$\measuredangle H(j\omega)=-\operatorname{arctg}\frac{4}{2-4}=-\operatorname{arctg}\frac{4}{-2}=1.107$$

! ona je dobila -2.034?

jer kalkulator računa kao da je u 4. kvadrantu pa moramo ODUZETI π

$$y_p(t)=3\cdot 0{,}2236\cdot\cos(2t-2.034)\ ?$$

$y''+2y'+2y=u(t) \rightarrow s^2+2s+2=0$

$s_{1,2}=\dots$ karakt frekv = polovi

Realan dio polova (svih) mora biti >0 da sustav bude stabilan

*ako sustav nije stabilan $\Rightarrow$ ne možemo odrediti frekv karakt.

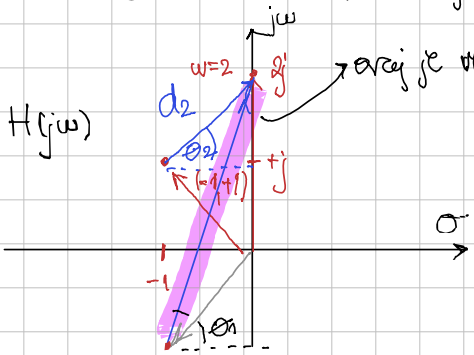
↳ to napišemo

↳ ne možemo odrediti odziv na $\cos(\omega t)$ (ona gore metoda)

Crtanje frekv. karakt

$$H(s)=\frac{1}{s^2+2s+2}=\frac{1}{(s-(-1-j))(s-(-1+j))}\Rightarrow\frac{1}{\underbrace{(j\omega-(-1-j))}_{d_1e^{j\theta_1}}\underbrace{(j\omega-(-1+j))}_{d_2e^{j\theta_2}}}$$

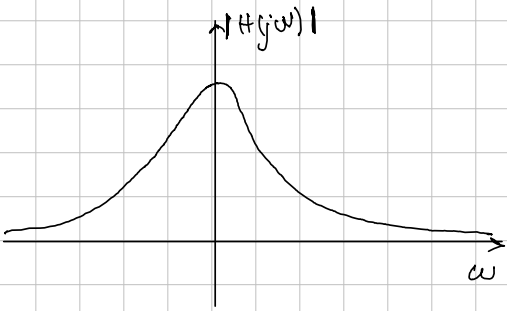
$H(j\omega)$



svaj je visok ovisno o ω

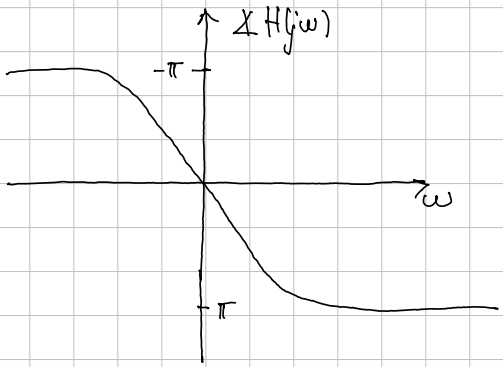
$$H(j\omega)=\frac{1}{d_1\cdot e^{j\theta_1}\cdot d_2e^{j\theta_2}}=\frac{1}{d_1\cdot d_2}e^{-j(\theta_1+\theta_2)}$$

$$|H(j\omega)|=\frac{1}{d_1\cdot d_2}$$



$\omega\uparrow \Rightarrow d_1\uparrow,\ d_2\uparrow \Rightarrow \frac{1}{d_1d_2}\downarrow$

pomičemo ω i gledamo kako se mijenjaju d_1 i d_2



$\measuredangle H(j\omega)=-(\theta_1+\theta_2)$

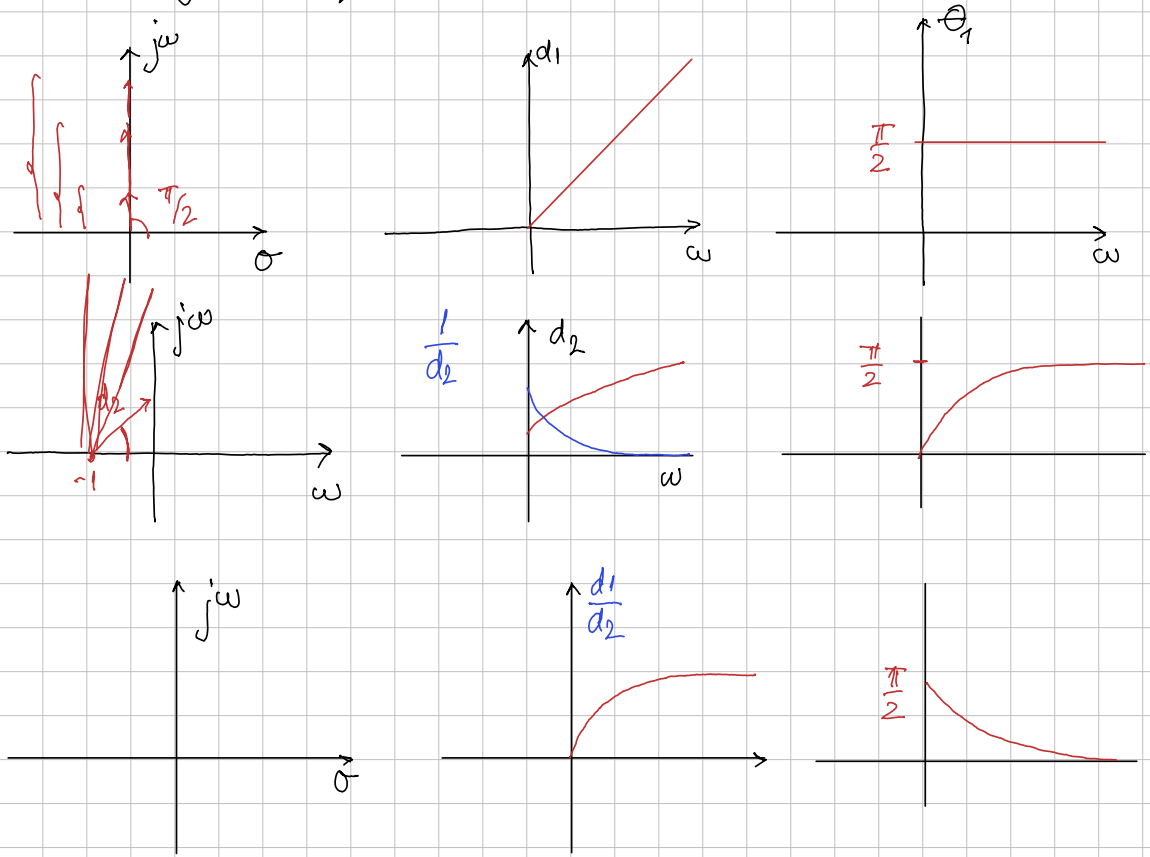
$\omega\uparrow\ \theta_2\to\frac{\pi}{2},\ \theta_1\to\frac{\pi}{2}$

teži $\omega\ -\pi$

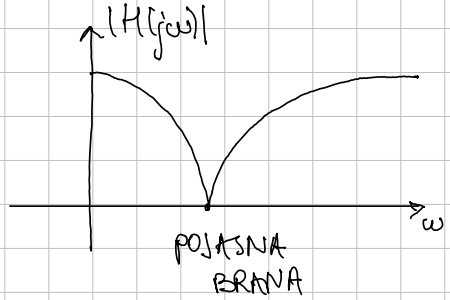
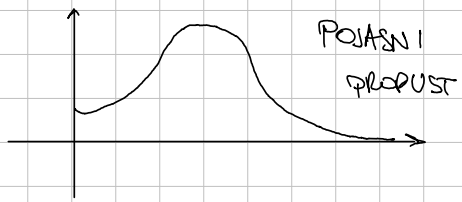
NISKOPROPUSNI F.

Primjer 2.) $H(s) = \dfrac{s}{s+1}$

$$H(j\omega) = \frac{j\omega}{j\omega+1} = \frac{d_1 e^{j\theta_1}}{d_2 e^{j\theta_2}} = \frac{d_1}{d_2} e^{j(\theta_1-\theta_2)}$$



VISOKOPROPUSNI F.



kada imamo $\emptyset$ u brojniku